INCH-POUND

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29 December 1989
SUPERSEDING
WS 21902
8 November 1983

MILITARY SPECIFICATION

INSULATION, CLOSURE, IN-PROCESS REQUIREMENTS FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the in-process requirements for one type of closure insulation, referred to herein as the insulation (see 6.1). The in-process requirements herein cover insulation characteristics which must be verified prior to assembly and fabrication of the end product.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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SPECIFICATIONS

MILITARY

MIL-D-3464 Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia Absorbent Gauze

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 In-process. The insulation shall conform to the requirements specified herein.

3.2 Surface condition. The surface condition of the insulation shall not exceed the permissible condition extent specified in table I.

TABLE I. Surface limits.

Condition 1/	Permissible Condition Extent
1. Blemishes (see 6.3.1)	Any blemish less than 1/2 square inch in area, 1 inch in length, or 0.030 inch deep None closer together than 1 inch. No more than five per surface.
2. Delaminations (see 6.3.2)	None allowed.
3. Cracks (see 6.3.3)	None allowed.
4. Wrinkled area (see 6.3.4)	None allowed.

1/ Any defects in overstock area (see 6.3.5) are acceptable (see figure 1).

3.3 Physical properties. Physical properties of the insulation shall be as specified in table II.

3.4 Workmanship. The insulation shall be free from surface contaminants, grease, oil, foreign material, and other defects which could render the part unsuitable for the intended use.

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TABLE II. Physical properties.

Property	Limits	
	Minimum	Maximum
Hardness, Barcol	65	85
Specific gravity 23/23 degrees Celsius (°C)	1.44	1.55
Residual volatiles, percent(%)	0.00	2.35
Resin content, %	32.50	41.00
Summation of resin content and residual volatiles, %	32.50	42.25
Compressive strength, pounds per square inch (psi), with ply	19,450	40,000

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table III.

TABLE III. Quality conformance inspection.

Inspection	Requirement	Inspection Method
Visual examination	3.2 & 3.4	4.3.1
Hardness test	3.3	4.3.2
Specific gravity test	3.3	4.3.3
Residual volatiles test	3.3	4.3.4
Resin content test	3.3	4.3.5
Summation of residual volatiles and resin content	3.3	4.3.6
Compressive strength test	3.3	4.3.7

4.2.1 Sampling for tests.

4.2.1.1 Lot. For inspection purposes, a lot shall consist of all the insulations which are manufactured with the same lot of material and cured in the autoclave at the same time and subject to inspection at the same time.

4.2.1.2 Selection of samples. Sample insulations shall be selected in accordance with table IV for the physical properties tests specified in 4.3.2 through 4.3.7. Specimens for the tests shall be obtained as specified in 4.2.2.

TABLE IV. Sampling for physical properties tests.

Number of Insulations in lot	Number of Sample Insulations to be Selected for Testing
1 - 6	1
7 - 12	2
13 - 18	3
19 - 24	4

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4.2.1.3 **Noncompliance of samples.** In the event that one or more of the samples fails to meet the requirements specified in 3.3, each insulation in the lot shall be subjected to the physical properties tests specified herein to determine compliance with requirements on an individual basis.

4.2.2 **Test rings and specimens.** The test ring shall be removed from the area shown on figure 1 for each insulation to be subjected to the physical properties tests specified herein. The ring shall be stored in a sealed bag and identified with the insulation's part number and serial number, and the serial numbers of the other insulations in the lot (see 4.2.1). Test specimens shall be removed from each ring and prepared for testing as indicated in the applicable test methods.

4.2.3 **Classification of characteristics.** The characteristics verified by the tests and examination herein are classified as Major or Minor in accordance with WR-43. Tests that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the test. The visual examination herein is not annotated with a classification code and serial number and is classified Minor.

4.3 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all inspections defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.3.1 **Visual examination.** Each insulation shall be visually examined to determine conformance to 3.2 and 3.4. Noncompliance with any specified requirements shall constitute cause for rejection. The following methyl ethyl ketone (MEK) solvent wipe method shall be used as an examination aid in determining conformance to 3.2.

a. Wet surface with MEK conforming to ASTM D 740.

b. Wipe off excess MEK with type VIII absorbent gauze conforming to The United States Pharmacopeia.

c. Allow part to dry in air. Defects will appear as well-defined wetted lines.

4.3.2 **Barcol hardness test.** (M102) Barcol hardness shall be determined using a Barcol hardness tester in accordance with the following:

a. Remove three test specimens (0.30 $\pm$ 0.06 inch by 0.30 $\pm$ 0.06 inch by 0.60 $\pm$ 0.06 inch) 120 $\pm$ 24° apart, from the test ring (see 4.2.2). Maximum surface roughness of the specimens shall be 125 microinches root mean square (RMS).

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b. Place test specimen on a hard flat surface.

c. Shim the Barcol hardness tester impressor legs to the same height as test specimen.

d. Grasp the impressor between legs and point sleeve, and set point sleeve above the point to be tested.

NOTE: Impressions must not be made within 1/8 inch of the specimen edge.

e. Quickly apply sufficient hand pressure (10 to 15 pounds) on the housing, and record the highest dial reading.

f. Repeat the above operation taking another four readings at random locations on that surface.

g. Turn the specimen over and repeat steps b through f.

h. Record the average of ten readings to nearest 1.0 unit.

i. Repeat steps b through h for each test specimen.

j. Calculate the average hardness of the three specimens. Noncompliance of the average reading with the Barcol hardness requirement specified in 3.3 shall be cause for rejection of the insulation.

k. Retain the specimens for the specific gravity test (see 4.3.3).

4.3.3 **Specific gravity test.** (M101) Specific gravity shall be determined in accordance with the following:

a. Obtain three specimens from the hardness tests (see 4.3.2).

b. Clean each specimen using clean cloth or laboratory towel moistened with MEK, and allow specimen to dry in air for a minimum of 20 minutes.

c. Weigh each specimen in air, and record weight to nearest 0.1 mg.

d. Weigh each specimen in kerosene, and record weight to nearest 0.1 mg.

e. Calculate specific gravity of each specimen as follows:

$$\text{Specific gravity at } 23/23\ ^{\circ}C = \frac{A \times C}{A - B}$$

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Where: A = weight of specimen in air, g

B = weight of specimen in kerosene, g

C = specific gravity of kerosene at 23/23°C.

f. Record specific gravity of each specimen to the nearest 0.01 unit.

g. Calculate the average specific gravity of the three specimens to the nearest 0.01 unit. Noncompliance of the average reading with the specific gravity requirement specified in 3.3 shall be cause for rejection of the insulation.

4.3.4 **Residual volatiles test.** (M103) Residual volatiles shall be determined in accordance with the following:

a. Remove three test specimens (0.30 ± 0.06 inch by 0.30 ± 0.06 inch by 0.60 ± 0.06 inch) 120 ± 24 degrees apart, from the test ring (see 4.2.2).

b. Clean each specimen using clean cloth or laboratory towel moistened with MEK, and allow specimen to dry in air for a minimum of 20 minutes.

c. Pre-dry specimens for minimum of 18 hours in desiccant conforming to MIL-D-3464.

d. Weigh each test specimen (to the nearest 0.001 gram (g)) in a tared crucible which has been previously heated at 325 ± 10 degrees Fahrenheit (°F) for minimum of one hour and cooled to room temperature in desiccator.

e. Place each crucible and specimen into air circulating oven which has been preheated to and maintained at 325 ± 10° F.

f. Heat crucibles and specimens in oven for 4 to 4.5 hours.

g. Remove the crucibles and specimens from the oven, and cool in a desiccator to room temperature.

h. Weigh each crucible and contents to nearest 0.001 g.

i. Calculate residual volatiles of each specimen as follows:

$$\text{Percent residual volatiles} = \frac{(A - B)}{A} (100)$$

Where: A = sample weight before heating, g

B = sample weight after heating, g

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j. Record average of three determinations to nearest 0.01 percent. Noncompliance of the average determination with the residual volatiles requirement specified in 3.3 shall be cause for rejection of the insulation.

k. Retain the three specimens for the resin content test (see 4.3.5).

4.3.5 Resin content test. (M104) Resin content shall be determined in accordance with the following:

a. Obtain test specimens from residual volatiles test (see 4.3.4).

b. Record the specimen weight to nearest 0.001 g.

c. Place the test specimen in Vycor tube and tightly attach the tube to test apparatus as shown on figure 2.

d. Start the vacuum pump, and evacuate the system. Check for air leaks by clamping off hose to vacuum pump and noting manometer. The leak rate must be less than 5 micrometers (0.0002 inch) Hg per minute. Recheck joints until this level is achieved.

e. With the vacuum on and pressure stabilized, place a Fisher burner under the specimen. Adjust flame area to completely cover sample.

NOTE: A tube furnace maintained at 1500 $\pm$ 25°F may be used in place of the Fisher burner.

f. Continue pyrolysis until manometer or gauge reading is same reading as before pyrolysis.

g. Turn off the burners, and cool under vacuum until the specimen is at room temperature (touch Vycor tube just under test specimen).

h. Turn off the vacuum, and carefully let air into the system. When pressure equilibrium is reached, remove the specimen from the test tube and weigh to the nearest 0.001 g.

i. Repeat b through h for each test specimen.

j. Calculate resin content of each specimen as follows:

$$\text{Dry resin content, percent} = (1.00 - \frac{A}{B})\,(100)K$$

Where: A = weight of specimen after pyrolysis, g

B = weight of specimen before pyrolysis, g

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K = constant ratio of percent resin (Soxhlet extraction or di-methyl-formamide (DMF) wash) to percent weight loss by pyrolysis (K - U.S. Polymeric = 2.008, K - Fiberite = 2.060).

k. Record resin content average of the three specimens to the nearest 0.01 percent. Noncompliance of the average reading with the resin content requirement specified in 3.3 shall be cause for rejection of the insulation.

4.3.6 **Summation of volatiles and resin content.** (M105) The summation of the results of the tests performed in 4.3.4 and 4.3.5 shall be examined for conformance to the summation of residual volatiles and resin content requirement specified in 3.3. Noncompliance with the requirement shall be cause for rejection of the insulation.

4.3.7 **Compressive strength test.** (M106) Compressive strength shall be determined in accordance with the following:

a. Remove three test specimens (0.300 ± 0.005 inch by 0.300 ± 0.005 inch by 0.600 ± 0.005 inch) 120 ± 24 degrees apart, from the test ring (see 4.2.2). Maximum surface roughness shall be 125 microinches RMS. Specimens shall be machined such that length of specimen (0.600 inch reference) is tangent (within ± 18 degrees) to test ring curvature.

b. Measure (to the nearest 0.001 inch) length, width, and height of all test specimens with a calibrated micrometer. Record all readings on a data sheet. Calculate and record cross-section of each specimen.

c. Set the test specimen on load cell table of an Instron tensile tester equipped with a Model CF load cell with the length (0.600 reference) normal to the faces of the tester.

d. Depress DOWN button to start compression. Cross-head speed should be 0.05 inch-per-minute.

e. When the test sample fails, push RETURN button. This will return crosshead to ZERO position.

f. Repeat c through e for each test specimen.

g. Calculate compressive of each specimen as follows:

$$\text{Compressive strength} = \frac{C}{A}$$

Where: C = maximum load, pounds

A = cross-sectional area, square inch

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h. Record the average compressive strength of the three specimens to the nearest psi. Noncompliance of the average reading with the compressive strength requirement specified in 3.3 shall be cause for rejection of the insulation.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The insulation specified herein is intended for use as a component in the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

6.3 **Definitions.**

6.3.1 **Blemishes.** Any superficial imperfection or marks or deformities such as scratches, crazing, gouges, cuts, or stains.

6.3.2 **Delamination.** A separation between fabric plies which has sharply defined boundaries that are parallel to the fabric plies.

6.3.3 **Crack.** A parting or separation which extends across the fabric plies.

6.3.4 **Wrinkled areas.** Distortions in fabric orientation normal to the plane of the fabric.

6.3.5 **Overstock area.** Excess material which will be removed at a higher level of assembly (see figure 1).

6.4 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust Mark 104
Standard Missile

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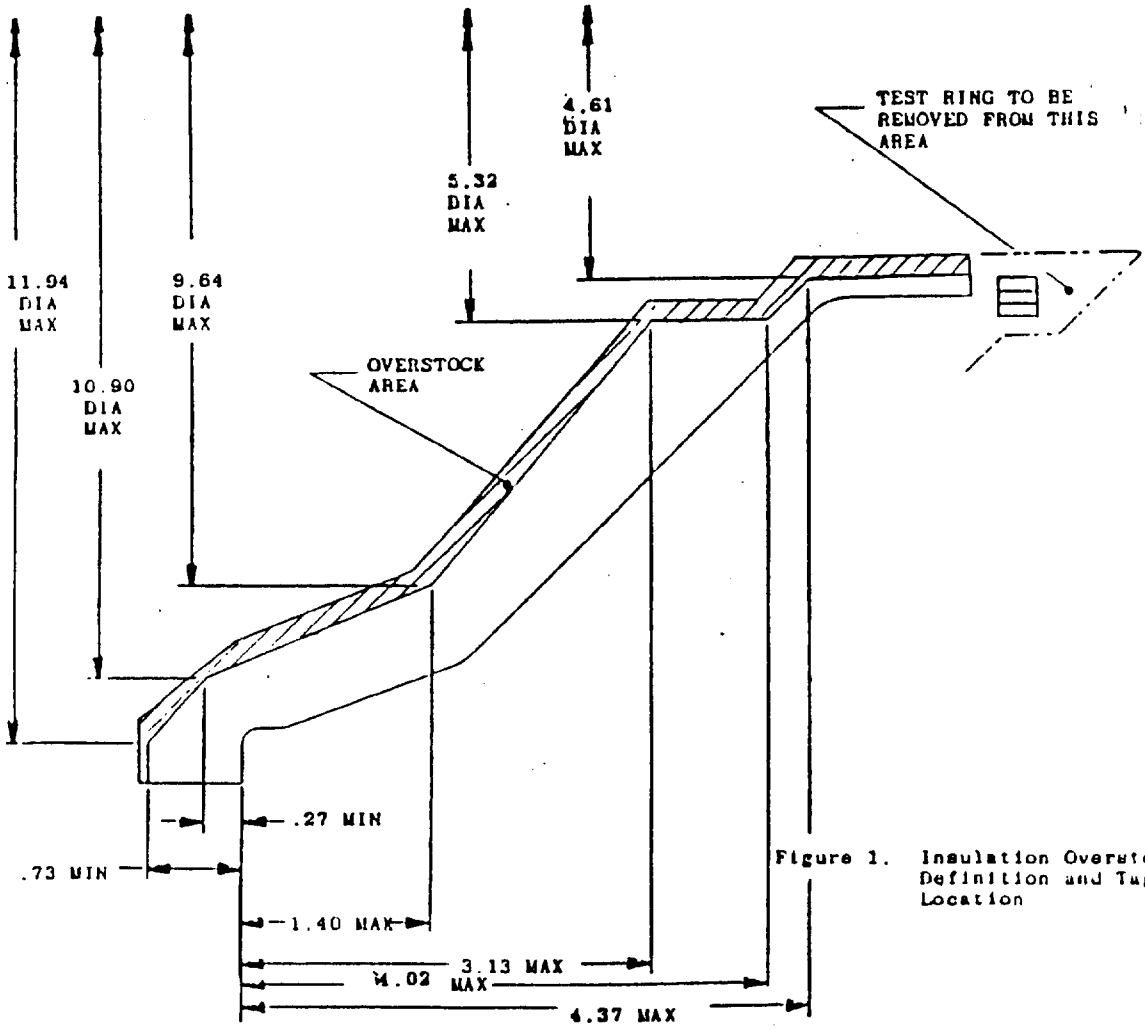


Figure 1. Insulation Overstock Definition and Tag End Location

FIGURE 1. Insulation overstock and tag end location.

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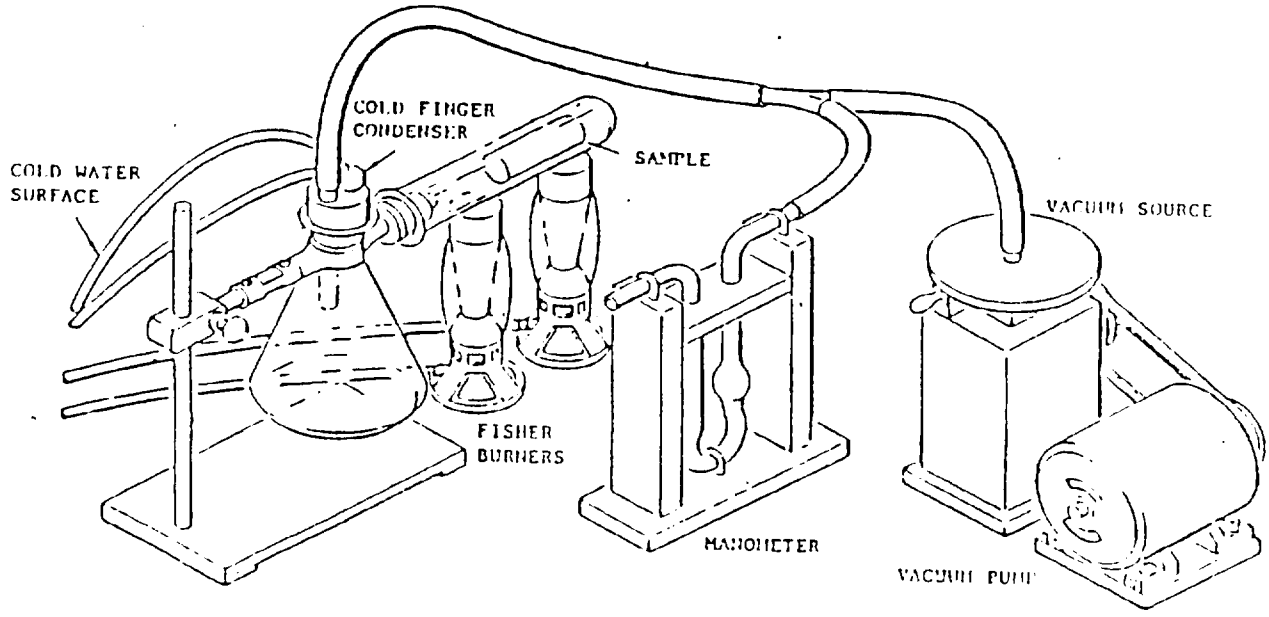


FIGURE 2. Resin content apparatus setup.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-82771	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
INSULATION, CLOSURE

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

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a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

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